

# Binary Classification of Obesity Risk Using Behavioral and Physical Attributes<sup>1</sup>

## Project Context

This project focuses on building an end-to-end predictive machine learning pipeline using structured health-related data.

The goal is to demonstrate proficiency in problem framing, model comparison, scaling, hyperparameter optimization, and model interpretation, aligned with applied analytics and Business Intelligence practices.

The dataset used is the UCI Estimation of Obesity Levels Based on Eating Habits and Physical Condition, which contains demographic, behavioral, and physical attributes for individuals, along with labeled obesity levels.

[UCI Repository](#)

## Problem Definition

Binary Classification Task

We reformulate the original multi-class obesity labels into a binary, decision-oriented target:

0 = No Obesity (Normal Weight + Insufficient Weight)

1 = Obesity Risk (Overweight + All Obesity categories)

This framing transforms the problem from academic categorization to a practical risk-identification task, suitable for real-world screening, intervention prioritization, or policy analysis.

## Model Selection — AutoSklearn<sup>2</sup>

Goal: Select the most appropriate classification model for a binary obesity-risk prediction problem using AutoSklearn2 and cross-validation.

This notebook focuses only on model selection. Hyperparameter optimization and final evaluation are handled in separate notebooks.

```
In [1]: from auto_sklearn2 import AutoSklearnClassifier

import pandas as pd
import numpy as np
```

```
from sklearn.model_selection import StratifiedKFold
from sklearn.metrics import roc_auc_score, f1_score
```

## Load Data

```
In [2]: import pandas as pd
from pathlib import Path

# Define project root dynamically
PROJECT_ROOT = Path.cwd()

# Path to data folder
DATA_PATH = PROJECT_ROOT / "Data"

# Load dataset
df = pd.read_csv(DATA_PATH / "ObesityDataSet_raw_and_data_synthetic.csv")

df.head()
```

```
Out[2]:
```

|   | Gender | Age  | Height | Weight | family_history_with_overweight | FAVC | FCVC | NCP |
|---|--------|------|--------|--------|--------------------------------|------|------|-----|
| 0 | Female | 21.0 | 1.62   | 64.0   | yes                            | no   | 2.0  | 3.0 |
| 1 | Female | 21.0 | 1.52   | 56.0   | yes                            | no   | 3.0  | 3.0 |
| 2 | Male   | 23.0 | 1.80   | 77.0   | yes                            | no   | 2.0  | 3.0 |
| 3 | Male   | 27.0 | 1.80   | 87.0   | no                             | no   | 3.0  | 3.0 |
| 4 | Male   | 22.0 | 1.78   | 89.8   | no                             | no   | 2.0  | 1.0 |

```
In [4]: # Binary target
df["obesity_risk"] = df["NObeyesdad"].apply(
    lambda x: 0 if x in ["Insufficient_Weight", "Normal_Weight"] else 1
)

y = df["obesity_risk"]

# Remove target original e variáveis com Leakage
X = df.drop(
    columns=[
        "NObeyesdad",
        "obesity_risk",
        "Height",
        "Weight"
    ]
)

X = X.reset_index(drop=True)
y = y.reset_index(drop=True)
```

```
In [5]: # Identify categorical columns
categorical_columns = X.select_dtypes(include="object").columns
categorical_columns
```

```
Out[5]: Index(['Gender', 'family_history_with_overweight', 'FAVC', 'CAEC', 'SMOKE',
              'SCC', 'CALC', 'MTRANS'],
              dtype='object')
```

```
In [6]: # One-Hot Encoding
X_encoded = pd.get_dummies(
    X,
    columns=categorical_columns,
    drop_first=True
)
```

```
In [7]: X_encoded.shape
```

```
Out[7]: (2111, 21)
```

```
In [8]: X = X.reset_index(drop=True)
y = y.reset_index(drop=True)

X.shape, y.shape
```

```
Out[8]: ((2111, 14), (2111,))
```

## Cross-Validation Setup

```
In [9]: k = 5

cv = StratifiedKFold(
    n_splits=k,
    shuffle=True,
    random_state=42
)
```

## AutoSklearn2 — Model Selection via CV

```
In [10]: cv_roc_auc = []
cv_f1 = []

for fold, (train_idx, val_idx) in enumerate(cv.split(X_encoded, y), start=1):

    X_train_cv = X_encoded.iloc[train_idx]
    X_val_cv = X_encoded.iloc[val_idx]
    y_train_cv = y.iloc[train_idx]
    y_val_cv = y.iloc[val_idx]

    automl = AutoSklearnClassifier(
        time_limit=120,
        random_state=42
    )

    automl.fit(X_train_cv, y_train_cv)

    y_val_pred = automl.predict(X_val_cv)
    y_val_proba = automl.predict_proba(X_val_cv)[:, 1]
```

```
cv_roc_auc.append(roc_auc_score(y_val_cv, y_val_proba))  
cv_f1.append(f1_score(y_val_cv, y_val_pred))
```



```

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Error evaluating minmax_scaler + qda:
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```
Error evaluating robust_scaler + multinomial_nb:
```

```
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```

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```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.py", line 794, in fit
```

```
    self._count(X, Y)
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.py", line 921, in _count
```

```
    check_non_negative(X, "MultinomialNB (input X)")
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validation.py", line 100, in check_non_negative
```

```
ion.py", line 1775, in check_non_negative
    raise ValueError(f"Negative values in data passed to {whom}.")
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Error evaluating standard\_scaler + multinomial\_nb:  
 All the 5 fits failed.  
 It is very likely that your model is misconfigured.  
 You can try to debug the error by setting error\_score='raise'.

Below are more details about the failures:

```

-----
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin

```



```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increas
e the value of `reg param` to reduce the collinearity.
```

Error evaluating minmax\_scaler + qda:  
All the 5 fits failed.  
It is very likely that your model is misconfigured.  
You can try to debug the error by setting error score='raise'.

Below are more details about the failures:

4 fits failed with the following error:

```
Traceback (most recent call last):
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increase the value of `reg_param` to reduce the collinearity.
```

```
1 fits failed with the following error:
```

Traceback (most recent call last):

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
```

```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increase
the value of `reg_param` to reduce the collinearity.

```

Error evaluating robust\_scaler + multinomial\_nb:  
All the 5 fits failed.  
It is very likely that your model is misconfigured.  
You can try to debug the error by setting error\_score='raise'.

Below are more details about the failures:

```

-----
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 794, in fit
    self._count(X, Y)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 921, in _count
    check_non_negative(X, "MultinomialNB (input X)")
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validat
ion.py", line 1775, in check_non_negative
    raise ValueError(f"Negative values in data passed to {whom}.")
ValueError: Negative values in data passed to MultinomialNB (input X).

```

Error evaluating robust\_scaler + qda:  
All the 5 fits failed.  
It is very likely that your model is misconfigured.  
You can try to debug the error by setting error\_score='raise'.

Below are more details about the failures:

```

-----
4 fits failed with the following error:
Traceback (most recent call last):

```

```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

```

-----

1 fits failed with the following error:

Traceback (most recent call last):

```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

```

Error evaluating standard\_scaler + multinomial\_nb:

All the 5 fits failed.

It is very likely that your model is misconfigured.

You can try to debug the error by setting error\_score='raise'.

Below are more details about the failures:

-----

5 fits failed with the following error:

Traceback (most recent call last):

```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin

```



```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
```

```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increase
the value of `reg_param` to reduce the collinearity.

```

Error evaluating robust\_scaler + multinomial\_nb:  
All the 5 fits failed.  
It is very likely that your model is misconfigured.  
You can try to debug the error by setting error\_score='raise'.

Below are more details about the failures:

```

-----
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line
1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 794, in fit
    self._count(X, Y)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.p
y", line 921, in _count
    check_non_negative(X, "MultinomialNB (input X)")
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validat
ion.py", line 1775, in check_non_negative
    raise ValueError(f"Negative values in data passed to {whom}.")
ValueError: Negative values in data passed to MultinomialNB (input X).

```

Error evaluating robust\_scaler + qda:  
All the 5 fits failed.  
It is very likely that your model is misconfigured.  
You can try to debug the error by setting error\_score='raise'.

Below are more details about the failures:

```

-----
4 fits failed with the following error:
Traceback (most recent call last):

```

```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

-----
1 fits failed with the following error:
Traceback (most recent call last):
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selecti
on_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py",
line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", lin
e 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_
analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increas
e the value of `reg_param` to reduce the collinearity.

```

## Cross-Validation Summary

```

In [11]: print("\n=== Cross-Validation Summary ===")
print(f"Mean ROC-AUC: {np.mean(cv_roc_auc):.4f}")
print(f"ROC-AUC Std: {np.std(cv_roc_auc):.4f}")
print(f"Mean F1: {np.mean(cv_f1):.4f}")

```

```

=== Cross-Validation Summary ===
Mean ROC-AUC: 0.9779
ROC-AUC Std: 0.0059
Mean F1: 0.9509

```

## Train Final AutoSklearn2 Model

```
In [14]: automl_final = AutoSklearnClassifier(  
        time_limit=120,  
        random_state=42  
    )  
  
    automl_final.fit(X_encoded, y)
```



```

File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increase the value of `reg_param` to reduce the collinearity.

-----
1 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
    raise linalg.LinAlgError(
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increase the value of `reg_param` to reduce the collinearity.

Error evaluating minmax_scaler + qda:
All the 5 fits failed.
It is very likely that your model is misconfigured.
You can try to debug the error by setting error_score='raise'.

```

Below are more details about the failures:

```

4 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn.pipeline.py", line 621, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
    raise linalg.LinAlgError(

```

```
numpy.linalg.LinAlgError: The covariance matrix of class 1 is not full rank. Increase the value of `reg_param` to reduce the collinearity.
```

```
-----
1 fits failed with the following error:
```

```
Traceback (most recent call last):
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
```

```
    estimator.fit(X_train, y_train, **fit_params)
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
```

```
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\discriminant_analysis.py", line 1132, in fit
```

```
    raise linalg.LinAlgError(
```

```
numpy.linalg.LinAlgError: The covariance matrix of class 0 is not full rank. Increase the value of `reg_param` to reduce the collinearity.
```

```
Error evaluating robust_scaler + multinomial_nb:
```

```
All the 5 fits failed.
```

```
It is very likely that your model is misconfigured.
```

```
You can try to debug the error by setting error_score='raise'.
```

```
Below are more details about the failures:
```

```
-----
5 fits failed with the following error:
```

```
Traceback (most recent call last):
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\model_selection\_validation.py", line 833, in _fit_and_score
```

```
    estimator.fit(X_train, y_train, **fit_params)
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\pipeline.py", line 621, in fit
```

```
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\base.py", line 1336, in wrapper
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.py", line 794, in fit
```

```
    self._count(X, Y)
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\naive_bayes.py", line 921, in _count
```

```
    check_non_negative(X, "MultinomialNB (input X)")
```

```
File "C:\Users\biauser\anaconda3\envs\edal\Lib\site-packages\sklearn\utils\validation.py", line 100, in check_non_negative
```



Out[14]: ▾ AutoSklearnClassifier ⓘ

► Parameters

```
In [20]: print("\n=== Cross-Validation Summary ===")
print(f"Mean ROC-AUC: {np.mean(cv_roc_auc):.4f}")
print(f"ROC-AUC Std: {np.std(cv_roc_auc):.4f}")
print(f"Mean F1: {np.mean(cv_f1):.4f}")
```

```
=== Cross-Validation Summary ===
Mean ROC-AUC: 0.9779
ROC-AUC Std: 0.0059
Mean F1: 0.9509
```

```
In [21]: print("\n=== Selected Model (AutoSklearn2) ===")
print(automl_final.best_params)
```

```
=== Selected Model (AutoSklearn2) ===
{'preprocessor': 'standard_scaler', 'classifier': 'random_forest'}
```

## Model Selection Statement

### Model Selection Decision

Using AutoSklearn2 with cross-validation, the framework selected a Random Forest classifier as the best-performing model for this task.

AutoSklearn2 identified Random Forest as the best-performing model. In the next step, I explored both Random Forest and Logistic Regression. While Random Forest offered higher predictive power, Logistic Regression provided better interpretability, which is important for risk-screening contexts.

In [ ]: